

NovaMart Data Science Project

Campaign ROI, Customer Segmentation, Lead Conversion, and Customer Growth

Detailed Project Brief

Client	NovaMart, a consumer retail brand operating across multiple product categories and digital channels
Role	You are the analytics team asked to convert messy operational data into decision-ready recommendations
Core theme	Marketing efficiency, customer growth, campaign effectiveness, and the quality of acquired customers
Submission expectation	Management wants defensible analysis, not dashboards without judgment

THIS IS AN OPEN-ENDED BUSINESS PROBLEM. THERE IS NO SINGLE CORRECT ROUTE, BUT THERE MUST BE A DEFENSIBLE ONE.

Executive mandate

How should NovaMart improve marketing efficiency and customer growth over the next quarter using evidence from campaign, lead, digital behavior, and transaction data?

Senior management believes three things may be happening at the same time:

Marketing spend is not being allocated efficiently across channels, regions, and campaign types.

Different customer groups respond very differently to promotions, content, devices, and campaign journeys.

The company is generating many leads, but too few become high-value repeat buyers.

Your job is not to report raw numbers. Your job is to build a trustworthy analytical foundation, discover what the data is actually saying, and recommend where management should act.

1. Business context

NovaMart is a consumer retail brand with a digital acquisition engine and a multi-category transaction business. The company runs campaigns across paid and owned channels, collects leads, observes website behavior, and records transactions after conversion.

Management has begun to question whether current marketing spending is producing the right kind of customer growth. Some campaigns may be generating volume but not value. Some audiences may be cheap to acquire but weak in long-term contribution. Some regions, devices, and content formats may be materially more effective than others.

You have been given raw operational data. The files contain inconsistencies, broken keys, duplicate values, malformed fields, missing observations, and category coding problems.

2. Data assets provided

You are being given multiple related files.

File	Rows	Primary business role	Important join field(s)	What may go wrong
customers	2,418	Customer profile and customer master	customer_id	Duplicate IDs, missing demographics, invalid ages, inconsistent labels
campaigns	39	Campaign setup, spend, reach, creative, objectives	campaign_id	Mixed date formats, malformed currency, inconsistent region labels, unusual clicks/impressions
leads	5,224	Lead generation and short-term conversion	lead_id; customer_id; campaign_id	Missing keys, inconsistent source labels, text-coded numerics, conversion timing issues

File	Rows	Primary business role	Important join field(s)	What may go wrong
website_sessions	19,045	Behavioral engagement and digital funnel signals	session_id; customer_id; campaign_id	Boolean coding issues, session quality anomalies, weak joins to campaign/customer
transactions	2,808	Commercial outcome, revenue, discounting, and return behavior	order_id; customer_id	Date inconsistencies, impossible values, discount issues, last-touch ambiguities

Expected relationships (do not assume perfect joins)

customers.customer_id ↔ leads.customer_id
 customers.customer_id ↔ website_sessions.customer_id
 customers.customer_id ↔ transactions.customer_id
 campaigns.campaign_id ↔ leads.campaign_id
 campaigns.campaign_id ↔ website_sessions.campaign_id

Scope of work

A. Data audit and analytical base table

Build a clean, analysis-ready foundation from the raw files.

Identify and document every major data quality issue that could materially affect business interpretation.

State and justify how you treated duplicates, missing values, inconsistent categories, malformed numerics, invalid dates, outliers, and failed joins.

Create business-ready KPIs that can be trusted by management. You may define additional metrics, but every metric must be clearly defined.

B. Descriptive and statistical diagnosis

Determine which channels, campaign objectives, creative types, regions, devices, and other key dimensions appear strongest on lead generation, conversion, revenue, repeat buying, and return on spend.

Assess whether observed differences in conversion, order value, revenue, or return behavior across important groups are large enough to be considered meaningful rather than accidental.

Examine whether discounting improves conversion and whether the effect varies by customer type, channel, region, campaign objective, or landing-page context.

Evaluate whether meaningful differences exist in customer value across regions, loyalty tiers, income bands, device preferences, or acquisition types.

Investigate which engagement signals from website sessions are most associated with lead conversion or purchasing.

Highlight which anomalies or untreated outliers materially distort the business story.

C. Customer segmentation

Identify distinct customer groups using a combination of demographic, behavioral, and commercial variables.

Profile each segment in plain business language rather than technical jargon.

For each segment, answer: who they are, how they behave, how valuable they appear, which campaigns or channels reach them best, and what management should do next quarter.

D. Campaign and channel grouping

Identify natural groupings of campaigns based on spend, reach, engagement, conversion, and commercial value outcomes.

Explain which campaign groups should be scaled up, redesigned, maintained, or stopped.

Separate high-volume campaigns from genuinely high-quality campaigns.

E. Lead conversion prediction

Develop a decision-ready approach to identify which leads are most likely to convert within 30 days.

Compare more than one predictive approach and defend the final choice on business and analytical grounds.

Explain which variables matter most and convert the model output into a prioritization rule that a business team could actually use.

F. Customer growth or retention prediction

Choose one of the following and solve it well: predict whether a newly acquired customer will make another purchase within 90 days, or predict whether a customer will become high value over the next period.

Explain which variables are most informative.

Identify early warning signs of weak-quality acquisition.

Recommend how targeting, discounting, and channel mix should change in response to the results.

G. Executive reporting layer

Create an executive-facing reporting solution that allows a non-technical marketing manager to understand what is working, what is underperforming, where money is being wasted, which customer groups deserve more budget, and which channels or campaigns deserve more budget next quarter.

The reporting layer should be usable by management, not only by analysts.

H. Final decision memo

Finish with a memo (Power Point with max 20 slides) to the CMO that states the top five findings, the top three actions for the next quarter, where budget should be increased, where budget should be reduced or redesigned, and what risks or data limitations leadership should know before acting.

Management questions that must be answered clearly

Which acquisition channels are generating the most leads, and which are generating the most valuable customers?

Which campaign types are efficient only on the surface because of volume, but weak after conversion, revenue, or repeat purchase are considered?

Which regions, cities, devices, or audience groups respond best to which marketing approaches?

Does discounting improve conversion enough to justify margin sacrifice, and for whom does it work best?

Which early digital behaviors signal serious intent versus low-quality traffic?

Which customer segments deserve differentiated messaging, budget, and retention treatment?

Can leads be prioritized more intelligently so commercial effort is focused where conversion probability is highest?

Where should management increase budget, where should it redesign the playbook, and where should it stop spending?

Required deliverables

Deliverable	What it must contain	Minimum standard
Data audit note	Documented data issues, treatment choices, assumptions, exclusions, and unresolved risks	Transparent and defensible
Clean analytical dataset / base table	Merged and prepared data foundation used for analysis	Usable and reproducible
Analytical report or slide deck	Business findings from descriptive, statistical, segmentation, and prediction work	Decision-oriented, not descriptive only
Executive reporting layer	Manager-friendly reporting view for performance monitoring	Easy to interpret
Final CMO memo	Top findings, actions, budget shifts, and limitations	Clear, prioritized, commercially sensible

7. Guardrails

You may not ignore data quality problems or bury them in an appendix.

You may not present metrics without defining them.

You must separate data issues from business findings.

You must explain assumptions and exclusions.

You must defend why your recommendations are financially and operationally sensible.

You should not treat model accuracy alone as business success; the recommendation must still make managerial sense.

You should not confuse high lead volume with high customer quality.

Important: This brief does not prescribe a tool, library, workflow, or sequence. The burden is on you to choose an appropriate approach and justify it.

Suggested structure of your submission

1. Data audit and data preparation logic
2. Analytical base table or integrated data model
3. Exploratory findings and metric definitions
4. Statistical findings and interpretation
5. Segmentation and managerial implications
6. Predictive modeling and operational scoring logic
7. Executive reporting layer
8. Final recommendations and limitations

9. What strong work will look like

A strong submission will combine technical rigor with business judgment.

It will show that the team can move from messy raw data to a coherent analytical narrative.

It will make thoughtful choices when data is imperfect rather than pretending the data is clean.

It will distinguish activity from effectiveness, and conversion from customer quality.

It will end with decisions that management could realistically act on in the next quarter.

Appendix: data-field examples

The following examples illustrate the nature of the raw fields. They are included to help you appreciate the variety of quality issues and business constructs present in the files.

Table	Illustrative fields	Why they matter
customers	gender, age, city, region, signup_date, preferred_device, loyalty_tier, income_band	Defines who the customer is and supports segmentation and value analysis
campaigns	channel, objective, dates, budget, spend, target_region, creative_type, impressions, clicks	Supports campaign efficiency, ROI, and grouping analysis
leads	lead_date, lead_source, landing_page, discount_offered_pct, lead_score, converted_30d, acquisition_cost_usd	Supports conversion diagnosis and lead prioritization
website_sessions	session_date, device, channel_group, pages_viewed, time_on_site_sec, bounce, add_to_cart, checkout_started	Captures digital intent and pre-purchase behavior
transactions	order_date, product_category, units, revenue_usd, discount_pct, payment_type, returned_flag, marketing_channel_last_touch	Captures realized value and downstream customer quality